

Unit 1

INFORMATION REPRESENTATION

59 items · 399 marks

Syllabus sections in this unit

- 1.1 Data Representation
- 1.2 Multimedia
- 1.3 Compression

Unit 1: Information representation

1. M/J 26/P11/Q1/a,b,c,d

1 A computer can store numbers and text in binary.

(a) The computer uses the UNICODE character set.

(i) Complete the table by calculating the missing values.

character	denary	binary	hexadecimal
⋮		0001 1110 1001	1E9
@	64		40
Ω	911	0011 1000 1111	
Ψ		0001 0101 1101 0110	15D6

[4]

working space

(ii) Identify **one** other character set and explain how it is different from UNICODE.

character set

differences

[3]

(b) The computer also uses two's complement.

Convert the 8-bit two's complement binary integer 01101011 into denary.

[1]

(c) Show how the denary number 30 is subtracted from the binary number 01101011 using binary subtraction.

Show your working.

[2]

(d) Identify **two** differences between binary prefixes and decimal prefixes.

1

2

[2]

2. M/J 26/P11/Q5/a,b,c

- 5** A school needs to create a new logo. The logo can be created as a bitmap image or a vector graphic.

(a) Explain how a bitmap image is encoded in a computer.

[3]

(b) Complete the description of a vector graphic by writing the missing terms.

A is an individual element of an image, for example: a line, a square.

A is an attribute of one part of the image, for example: a line colour, the width of a line.

A is all of the elements that are included in the vector graphic.

[3]

(c) Tick (✓) **one** box to identify whether a bitmap image or a vector graphic will be most appropriate for the school logo.

Justify your choice.

	tick (✓)
bitmap image	
vector graphic	

justification

[3]

3. M/J 26/P12/Q1/a,b,c,d

- 1 (a) Tick (✓) **one** box to identify the correct definition of a pixel in a bitmap image.

- | | |
|--------------------------|---|
| ☐ | a colour in an image |
| ☐ | a row of consecutive identical colours in an image |
| ☐ | the binary code for one colour in an image |
| ☐ | the smallest element of an image that can be one colour |

[1]

- (b) Describe the difference between the terms image resolution and screen resolution.

[2]

- (c) Two items in a bitmap image file header are the resolution and colour depth.

Identify **one** other item included in the file header of a bitmap image.

[1]

- (d) The following bitmap image is created. Each letter represents a colour: R is red, W is white, B is blue.

R	R	W	W	W	W
B	B	R	R	W	W
W	W	W	R	R	R
W	W	W	W	R	R
W	W	R	R	R	R
R	R	R	R	R	B
B	B	B	B	B	B

- (i) Identify the smallest bit depth that can be used for the image shown.

[1]

- (ii) Explain how the image shown can be compressed using lossless compression.

[3]

4. M/J 26/P12/Q4/a,b

4 A sound file has a sampling rate of 44.1 kHz.

(a) Describe what is meant by a sampling rate of 44.1 kHz.

[1]

(b) A student creates a recording of a sound with a sampling resolution of 2 bytes.

The student creates a second recording of the same sound but with a sampling resolution of 8 bits.

Explain how the two sound recordings **and** the two sound files will be affected by the different sampling resolutions.

[3]

5. M/J 26/P12/Q5/a,b

5 (a) Complete the table by calculating the missing values.

denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)
160	10100000	A0	
	100100001	121	001010001001
2486		9B6	0010010010000110
35123	1000100100110011		00110101000100100011

[4]

working space

(b) Complete the table by writing the answer to each statement.

statement	answer
The smallest 12-bit two's complement binary number	
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	

[3]

6. M/J 26/P13/Q3/a,b,c(i)

3 An animation is a sequence of bitmap images with a sound recording played at the same time.

(a) The sound is recorded with a sampling resolution of 2 bytes.

State what is meant by a sampling resolution of 2 bytes.

[1]

(b) (i) Explain how the data for the bitmapped images are encoded.

[3]

(ii) The bitmap images have a bit depth of 2 bytes. Each image is 2048 pixels wide by 1536 pixels high. One second of video contains eight individual bitmap images.

There is no sound recording in this part of the video.

Calculate an estimate for the file size of one second of the video in mebibytes (MiB). Show all of your working.

working:

answer MiB

[2]

(c) The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

(i) Explain the different ways that lossless compression can be used to compress the images and the sound file.

image file

sound file

[4]

7. M/J 26/P13/Q6/a,b,c,d

- 6 (a)** Draw **one or more** lines to match each denary number with its equivalent unsigned binary, hexadecimal or Binary Coded Decimal (BCD) value(s).

Denary number	Equivalent unsigned value
	Hexadecimal: 121
3120	BCD: 00010101100001010001
289	Binary: 110000110000
22516	Hexadecimal: 57F4
15851	Binary: 11110111101011
	BCD: 0011000100100000

[4]

- (b)** Complete the binary addition. Show all of your working.

$$\begin{array}{r}
 110101001111 \\
 + 111111001111 \\
 \hline
 \end{array}$$

[2]

- (c)** Convert the two's complement integer 011010110001 into denary.

[1]

- (d)** Tick (✓) **one** box to identify the smallest integer that can be represented by a 16-bit two's complement integer.

- ☐ 0000000000000000
☐ 1111111111111111
☐ 0111111111111110
☐ 1000000000000000

[1]

8. O/N 25/P11/Q1/a,b

- 1 (a) (i) Convert the binary number into hexadecimal.

101100111010

[1]

- (ii) Convert the denary number into Binary Coded Decimal (BCD).

108

[1]

- (iii) Convert the 12-bit two's complement binary integer into denary.

Show your working.

111110111100

Working

Denary value

[2]

- (b) (i) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ +\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0 \\ \hline \end{array}$$

[1]

- (ii) Name and describe the error that can occur when binary addition is performed.

[2]

9. O/N 25/P11/Q7/d

- 7 Web browsers use Internet Protocol (IP) addresses.

- (d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

- (i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working

File size of bitmap kibibytes

[2]

- (ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

[2]

- (iii) Explain how a bitmap image is compressed using run-length encoding (RLE).

[2]

10. O/N 25/P12/Q6/a,b,d

- 6 (a)** A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression

Justification

[1]

- (b)** The following table shows some words and corresponding denary values.

Word	Denary value
Computing	55
Science	56
Computers	57
are	58
Brilliant!	59
is	60
Fun!	61
Amazing!	62

The following table shows three bytes of data that have been received.

Use the table to find the corresponding words from the binary values received.

Binary value	00111000	00111100	00111110
Word			

Working

[1]

- (d)** A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

File size megabytes

[2]

11. O/N 25/P12/Q7/a,b,c

- 7 (a) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0\ 1\ 1\ 1 \\ +\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$

[1]

- (b) A computer uses the Unicode character set.

State the number of bits used to store **one** character from the Unicode character set.

[1]

- (c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

- (i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

[1]

- (ii) Write the hexadecimal value for the ASCII character 'h'.

[1]

12. O/N 25/P13/Q1/a,b,c

- 1 (a) A student creates sound recordings.

Making changes to a sound recording may impact the accuracy of the resulting sound file or the size of the file.

Draw **two** lines from each change to the impacts it has on the sound file.

Change	Impact
increase the duration of the recording	the file size gets bigger
	no change in the file size
increase the sampling rate	the file size gets smaller
	the accuracy of the sound file improves
decrease the sampling resolution	no change to the accuracy of the sound file
	the accuracy of the sound file worsens

[3]

- (b) The file names of the sound recordings are stored using the ASCII character set.

Explain how text is represented by the ASCII character set.

[2]

- (c) Give **two** differences between the ASCII and UNICODE character sets.

1

2

[2]

13. O/N 25/P13/Q2/a,b,d

- 2 (a)** Data in a computer system is represented in binary.

Put **one** tick (✓) in each row to identify the minimum number of bits used to store each example of data.

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139							
16 000 000 unique amplitude values							
an IPv4 address							
256 unique colours							
an IPv6 address							
the denary value 65 000							

[3]

- (b)** Convert the denary number –108 into a 12-bit two's complement binary number.

[1]

- (d)** Convert the following positive binary integer into hexadecimal.

1110001100111011

[1]

14. O/N 25/P13/Q7/b

- 7** A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (b)** A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

File size gigabytes

[2]

15. M/J 25/P11/Q2/b

- 2** Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

- (b) (i)** Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

[4]

- (ii)** Identify whether the lossy or lossless compression method is more appropriate for real-time bit streaming. Justify your answer.

Compression method

Justification

[3]

16. M/J 25/P11/Q3/a,b

3 A computer stores images and text files.

(a) One of the images is a bitmapped image.

Complete the table by writing the answer for each statement.

Statement	Answer
the term for the smallest element that makes up an image	
the largest number of different colours that can be represented with a bit depth of 8 bits	
the term for the dots per inch (dpi) when an image is displayed	

[3]

(b) The text in the files is stored using the Unicode character set.

(i) Give **two** advantages of using the Unicode character set instead of using the ASCII character set.

1

2

[2]

(ii) The Unicode character 'Ƨ' has the binary value: 0010 0111 0110 1110

Convert the binary value for the character 'Ƨ' into denary.

[1]

(iii) The Unicode character 'Σ' has the hexadecimal value 2140

Convert the hexadecimal code for the character 'Σ' into denary.

[1]

17. M/J 25/P12/Q2/a,b,c

- 2 (a) Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary

Hexadecimal

[2]

- (b) (i) Convert the two's complement binary integer 11100010 into denary.

[1]

- (ii) Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest

Largest

[2]

- (c) Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application

Justification

[2]

18. M/J 25/P13/Q1/a,b,c

- 1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

- (i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space

Answer MB

[2]

- (ii) The student takes a second photograph with a lower bit depth.

Explain the effect of decreasing the bit depth on the image and on the image file.

Effect on image

Effect on image file

[4]

- (b) A teacher records an audio file of the student playing the piano during a music lesson.

Complete the table by giving the term for each description about sound representation on a computer.

Description	Term
the number of times the amplitude is measured per time interval	
the number of bits used to store each amplitude measurement	
the type of sound wave before it is recorded by a computer	

[3]

(c) The student types a report on a computer in a history lesson.

(i) The computer uses the Unicode character set.

Give **two** characteristics of the Unicode character set.

1

2

[2]

(ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

19. M/J 25/P13/Q7/b,c

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r}
 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\
 + \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \\
 \hline
 \end{array}$$

[1]

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

20. O/N 24/P11/Q1/a,b,c,d

- 1 (a) State **one** difference between a tebibyte and a gigabyte.

[1]

- (b) (i) Convert the unsigned binary integer into hexadecimal.

110001100111

Answer

[1]

- (ii) Convert the two's complement binary number into denary.

100110010111

Answer

[1]

- (iii) Convert the Binary Coded Decimal (BCD) into denary.

010101110011

Answer

[1]

- (c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

Answer

[2]

- (d) State **one** reason why binary addition and subtraction can result in overflow.

[1]

21. O/N 24/P11/Q4/b

- 4 A student uses a laptop to write a program that is saved as a text file.

- (b) The student compresses the file before it is emailed to their teacher as an attachment.

- (i) Explain the benefits to the teacher of the attachment being a compressed file.

[3]

- (ii) Describe **one** lossless method of compressing a text file.

[3]

22. O/N 24/P12/Q7/a,b

7 A student takes a photograph of a science experiment.

(a) The photograph is saved as a bitmapped image.

(i) Define the following bitmap terms.

Colour depth

File header

[2]

(ii) Explain why changing the image resolution will affect the image quality and file size.

Image quality

File size

[2]

(iii) Identify **one** lossless method of compressing an image.

[1]

(b) The student draws a picture on paper that is scanned into the computer and saved as a vector graphic.

Define the vector graphic terms property and drawing list.

Property

Drawing list

[2]

23. O/N 24/P13/Q6/a,b,c

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set

Description

[2]

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(i) State the largest number of different colours that can be represented by 8 bits.

[1]

(ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

[1]

(iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

[2]

(c) (i) Explain how an analogue sound wave is converted into digital data.

[2]

(ii) Describe **one** method of compressing a sound file using lossy compression.

[2]

24. O/N 24/P13/Q8/a,b

8 (a) Convert the hexadecimal number 1FAB into denary.

Working

Denary value

[1]

(b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

Denary value

[3]

25. M/J 24/P11/Q2/d

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:

- The digital camera in the doorbell starts recording a video.
- A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

[3]

26. M/J 24/P11/Q7

- 7 Complete the binary addition. Show your working.

$$\begin{array}{r}
 10011110 \\
 01100001 \\
 + 00011001 \\
 \hline
 \end{array}$$

[3]

27. M/J 24/P12/Q2/d

- 2** A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

(i) Describe how the data for a bitmapped image is encoded.

[3]

(ii) Describe the contents of a vector graphic drawing list.

[2]

(iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

1

2

[2]

28. M/J 24/P12/Q7/a,b,c

- 7** A computer stores binary data.

(a) Tick (✓) **one** box only to identify the **largest** file size.

☐

3300 kibibytes

☐

0.3 megabytes

☐

3 mebibytes

☐

3300 kilobytes

[1]

(b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

Answer

[3]

(c) Convert the hexadecimal number C0F into denary.

Show your working.

Working

Answer

[2]

29. M/J 24/P13/Q1/a,b,c,d

- 1 (a) Complete the following description.

A kibibyte has a prefix. Three kibibytes is the same as bytes.

A megabyte has a prefix. Two terabytes is the same as gigabytes.

[4]

- (b) Convert the denary number 241 to hexadecimal.

[1]

- (c) State what is meant by an **overflow in binary addition**.

[1]

- (d) Computers use character sets when representing characters in binary.

- (i) Complete the table by identifying the number of bits each of the character sets allocates to each character.

Character set	Number of bits
ASCII	
extended ASCII	
Unicode	

[1]

- (ii) Explain how the word 'Clock' is represented by a character set.

[2]



30. M/J 24/P13/Q2/a,b,c

2 A photograph is stored as a bitmap image.

- (a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working

Answer megabytes

[2]

- (b) The photograph is compressed before being uploaded to a web server.

- (i) Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

1

2

3

[3]

- (ii) Explain how run-length encoding (RLE) will compress the photograph.

[2]

- (c) Identify **two** elements of a bitmap image that can be changed to reduce its file size.

1

2

[2]



31. O/N 23/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each vector graphic term to its most appropriate description.

Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

[2]

- (b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

Explanation

[3]

- (c) Explain why a bitmap image is often compressed before it is attached to an email.

[2]

32. O/N 23/P12/Q3/a,b,c

- 3 (a) State **one** difference between a kibibyte and a megabyte.

[1]

- (b) (i) Convert the denary value into a 12-bit two's complement binary integer.

–196

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer

[1]

- (iii) Convert the unsigned binary integer into denary.

000111010110

Answer

[1]

- (c) Identify **one** practical application of BCD **and** justify why BCD is used in this application.

Application

Justification

[2]

33. O/N 23/P12/Q6/a,b,c

- 6 (a) A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

Lossy	Lossless
☐	☐

Justification

[3]

- (b) Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

[3]

- (c) A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working

Estimated file size in mebibytes

[2]

34. O/N 23/P13/Q1/a,b

- 1 (a) State what is meant by **analogue data**.

[1]

- (b) Draw **one** line from each term to its most appropriate description.

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

[2]

35. O/N 23/P13/Q2/a,b

- 2 (a) Describe the impact of increasing the image resolution on the quality of a bitmap graphic.

[2]

- (b) Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working

Answer in kibibytes

[2]

36. O/N 23/P13/Q5/b(i)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

- (b) The program is distributed by downloading the source code and its library files from a web server.

- (i) Explain the reasons for compressing the files.

[2]

37. M/J 23/P11/Q1/a,b,c

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

(a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

[3]

(b) The bitmap images are photographs of the holiday locations.

- (i) Colour depth and image resolution are both included in the file header of a bitmap image.

Identify **two other** items that could be included in the file header of each photograph.

1

2

[2]

- (ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working

File size MB

[2]

(c) The photographs are compressed before they are uploaded to a web server. Customers download the photographs from this web server.

- (i) Explain the reasons why compressing the photographs will benefit the customers.

[3]

- (ii) An image can be compressed using run-length encoding (RLE).

Explain the reasons why RLE may **not** reduce the file size of a bitmap image.
Give **one** example in your answer.

[3]

38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

- 3** A computer has an Operating System (OS).

- (d) The computer stores data in binary form.

- (i) State the difference between a kibibyte and a kilobyte.

[1]

- (ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

[1]

- (iii) Convert the positive binary integer 11110010 into hexadecimal.

[1]

- (iv) Give the smallest **and** largest two's complement binary number that can be represented using 8 bits.

Smallest

Largest

[2]

- (v) Add the following two binary integers using binary addition. Show your working.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0 \\ +\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

[2]

39. M/J 23/P12/Q4/a,b,c

4 Data in a computer is stored in binary form.

(a) State the number of unique binary values that can be represented in 16 bits.

[1]

(b) Give the 8-bit one's complement representation of the denary number -120 .

Show your working.

Working

Answer

[2]

(c) Convert the hexadecimal number A04 into denary.

Show your working.

Working

Answer

[2]

40. M/J 23/P13/Q3/c

3 A mobile telephone is used to record a video.

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

[3]

(ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate

Sampling resolution

[4]

41. M/J 23/P13/Q7/a

7 A computer stores data in binary form.

(a) Draw **one** line from each description to its matching denary value.

Description	Denary value
	-127
The smallest integer that can be represented in 8-bit two's complement.	127
	-255
The largest integer that can be represented in 8-bit two's complement.	-128
	-256
The largest unsigned integer that can be represented in 8 bits.	256
	128
	255

[3]

42. O/N 22/P11/Q1/a,b,c,d

- 1 (a) (i) Convert the unsigned binary integer into denary.

00100111

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

00100111

Answer

[1]

- (iii) Convert the 8-bit two's complement binary integer into denary.

11100111

Answer

[1]

- (b) Perform the following binary subtraction. Show your working.

$$\begin{array}{r} 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \\ - \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\ \hline \end{array}$$

[2]

- (c) Give **one** similarity and **two** differences between the ASCII and Unicode character sets.

Similarity

Difference 1

Difference 2

[3]

- (d) Sound samples are recorded and saved in a file.

- (i) State what is meant by **sampling rate**.

[1]

- (ii) Explain the effect of increasing the **sampling resolution** on the sound file.

[2]

43. O/N 22/P11/Q7/c

- (c) An Operating System may include a utility program to compress text files.

Describe **one** appropriate method of compressing a text file.

[3]

44. O/N 22/P12/Q2/a,b

- 2 (a) (i) Convert the two's complement binary integer into denary.

10010110

Answer [1]

- (ii) Convert the unsigned binary integer into hexadecimal.

10010110

Answer [1]

- (iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

Answer [2]

- (b) Perform the following binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 1\ 1\ 0\ 0 \\ +\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array}$$

[1]

45. O/N 22/P12/Q6/b

- 6 (a) A student uses a networked laptop computer to send an email to a colleague.

- (b) The student records a sound file.

- (i) Explain the effect of increasing the sampling rate on the accuracy of the sound recording.

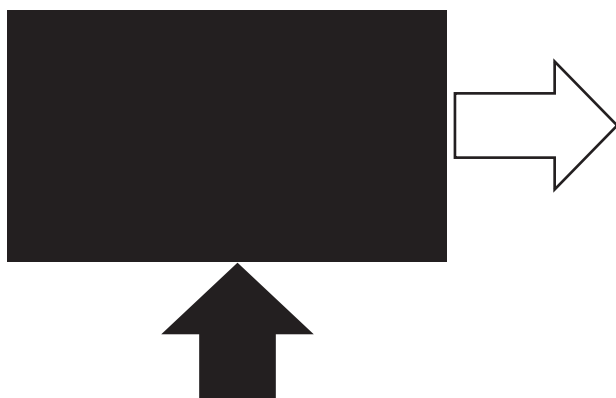
[2]

- (ii) Explain the effect of decreasing the sampling resolution on the file size of the sound recording.

[2]

46. O/N 22/P12/Q8/a,b,c

- 8 The following bitmap image has a resolution of 4096×4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920×1080 pixels.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the image file size.

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.			
Change the screen resolution to 1366×768 pixels.			
Change the colour of the rectangle from black to red.			

[2]

(b) State **two** benefits of creating a vector graphic instead of a bitmap image.

1

2

[2]

(c) A second bitmap image is stored using a colour depth of 8 bits per pixel.

The file is compressed using run-length encoding (RLE).

(i) The table shows the compressed and uncompressed values for parts of the image file.

Each colour of the pixel in the image is represented by a hexadecimal value.

Complete the table. The first row has been completed for you.

Uncompressed image	RLE compressed image
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
	2AB 2FF 11D 167
32 32 80 81 81	

[2]

(ii) RLE is an example of lossless compression.

Explain why lossless compression is more appropriate than lossy compression for a text file.

[2]

47. O/N 22/P13/Q1/a,b

1 A digital audio message needs to be recorded.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the accuracy of the recording.

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.			
Change the duration of the recording from 20 minutes to 40 minutes.			
Change the sampling resolution from 24 bits to 16 bits.			

[2]

(b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working

Answer megabytes

[2]

48. O/N 22/P13/Q2/e

- 2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (e) The photographs are stored as bitmap images.

Complete the statements about bitmap images by writing the missing words.

The of a bitmap image is the number of bits that are used to store each pixel.

Metadata about the image is stored in the of the file.

[2]

49. O/N 22/P13/Q9/a,b

- 9 (a) (i) Convert the unsigned binary value into hexadecimal.

10010011

Answer

[1]

- (ii) Convert the unsigned binary value into denary.

10010011

Answer

[1]

- (b) State **two** benefits of using Binary Coded Decimal (BCD) to represent values.

Benefit 1

Benefit 2

[2]

50. O/N 21/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

51. O/N 21/P11/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

52. O/N 21/P12/Q4/a,b,c,d,e

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(a) The binary value in the register represents an unsigned binary integer.

Convert the unsigned binary integer into denary.

[1]

(b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

[1]

(c) The binary value in the register represents a hexadecimal number.

Convert the binary number into hexadecimal.

[1]

(d) State why the value in the register cannot be interpreted as a Binary Coded Decimal (BCD).

[1]

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(i) Add the contents of **Register 1** and **Register 2**. Show your working.

Answer

[2]

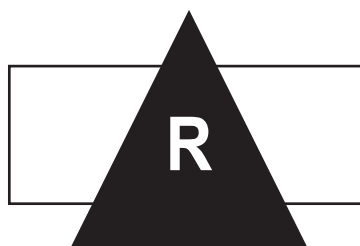
(ii) Subtract the contents of **Register 2** from the contents of **Register 1**. Show your working.

Answer

[2]

53. O/N 21/P12/Q5/a,b

5 Riya has created the following logo as a vector graphic.



- (a) Complete the table by writing a description of each vector graphic term **and** give an example for this logo.

Term	Description	Example from logo
Property		
Drawing list		

[4]

- (b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

- (i) Describe **two** differences between a vector graphic and a bitmap image.

1

2

[4]

- (ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1

Method 2

[4]

54. O/N 21/P13/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

55. O/N 21/P13/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

56. M/J 21/P11/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

- (b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

- (c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

- (i) Convert the hexadecimal code FC into denary.

[1]

- (ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

57. M/J 21/P12/Q4/b(i)

- 4 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system
# denotes a denary number, e.g. #123		

The current contents of the main memory and selected values from the ASCII character set are:

Address	Instruction
70	IN
71	CMP 100
72	JPE 80
73	CMP 101
74	JPE 76
75	JMP 80
76	LDD 102
77	INC ACC
78	STO 102
79	JMP 70
80	LDD 102
81	DEC ACC
82	STO 102
83	JMP 70
...	
100	68
101	65
102	100

ASCII code table (selected codes only)

ASCII code	Character
65	A
66	B
67	C
68	D

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

(i) The contents of memory address 300 represent an unsigned binary integer.

Write the denary value of the unsigned binary integer in memory address 300.

[1]

58. M/J 21/P12/Q6/a,b,c

6 A computer uses the ASCII character set.

(a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII

Extended ASCII

[2]

(b) Explain how a word such as 'HOUSE' is represented by the ASCII character set.

[2]

(c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

(i) Write the hexadecimal value for the Unicode character '1'.

[1]

(ii) Write the denary value for the Unicode character '5'.

[1]

59. M/J 21/P13/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

(i) Convert the hexadecimal code FC into denary.

[1]

(ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

Answers Section

1. M/J 26/P11/Q1/a,b,c,d

1(a)(i)	1 mark for each correct value <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>Character</th><th>Denary</th><th>Binary</th><th>Hexadecimal</th></tr> </thead> <tbody> <tr> <td>☼</td><td>489</td><td>0001 1110 1001</td><td>1E9</td></tr> <tr> <td>@</td><td>64</td><td>0100 0000</td><td>40</td></tr> <tr> <td>Ω</td><td>911</td><td>0011 1000 1111</td><td>38F</td></tr> <tr> <td>Ψ</td><td>5590</td><td>0001 0101 1101 0110</td><td>15D6</td></tr> </tbody> </table>	Character	Denary	Binary	Hexadecimal	☼	489	0001 1110 1001	1E9	@	64	0100 0000	40	Ω	911	0011 1000 1111	38F	Ψ	5590	0001 0101 1101 0110	15D6	4	Correct answers only
Character	Denary	Binary	Hexadecimal																				
☼	489	0001 1110 1001	1E9																				
@	64	0100 0000	40																				
Ω	911	0011 1000 1111	38F																				
Ψ	5590	0001 0101 1101 0110	15D6																				
1(a)(ii)	1 mark for the identification - ASCII 1 mark for each difference to max 2 - There are fewer bits per character // ASCII has 7/8 bits per character, UNICODE has many more - Fewer different characters can be represented // ASCII only represents the Latin alphabet - ASCII Cannot be expanded indefinitely	3	Accept extended ASCII, or any other valid character set. English Language BOD																				
1(b)	107	1	Correct answer only																				
1(c)	1 mark for correct working - showing/working binary subtraction 1 mark for correct answer 0100 1101 Method 1: Change +30 to -30 and add 30 in binary: 0001 1110 Flip bits: 1110 0001 +0000 0001 Add the 1: 1110 0010 Add -30: 0110 1011 1110 0010 (1) 0100 1101 Method 2: Direct Subtraction 0110 1011 -0001 1110 0100 1101 11 1 Carries	2	No marks for converting to denary and back Method 1: Direct subtraction Needs to show the carries for the working mark Method 2: Change +30 to -30 and add Needs to show the conversion and the addition																				
1(d)	1 mark each - Binary is in powers of 2 whilst decimal is in powers of 10 - Any example of values e.g. Kibibyte is 1024 bytes whilst Kilobyte is 1000 bytes	2	Do not allow: Decimal uses the digits 0,1,2,3,4,5,6,7,8,9 binary only uses the digits 0 and 1 The question asks about binary and denary prefixes MP2: Binary is mebibyte, denary is megabyte is ok, binary is mebibyte, denary is kilobyte is not																				

2. M/J 26/P11/Q5/a,b,c

5(a)	1 mark for each point to max 3 - The image is made of pixels - Each pixel has one colour - Each colour has a unique binary code - The code for the colour of each pixel is stored in sequence / order	3	
------	---	---	--

5(b)	1 mark for each correct term - drawing object - property - drawing list A drawing object is an individual element of an image, for example: a line, a square. A property is an attribute of one part of the image, for example: a line colour, the width of a line. A drawing list is all of the elements that are included in the vector graphic.	3	
5(c)	No mark for the choice. 1 mark for each point for a matching justification to max 3 Bitmap image: - The logo might include something like a photograph - Bitmaps can be more detailed/precise/high quality - which may be more attractive/appealing on documents - Bitmaps can be edited more precisely - in case changes need to be made to the logo in future Vector graphic: - A logo should be made of simple shapes which are calculated mathematically - A vector graphic logo can be resized without pixelation - so, it could be used on different documents without losing quality - The file size is smaller than a bitmap image - so, the logo will not use excessive file storage if used in many documents	3	Justification must match the candidate's choice. No mix and match. Accept any sensible justification of their choice Accept other examples such as tee shirts for documents in MP3 and MP8

3. M/J 26/P12/Q1/a,b,c,d

1(a)	1 mark for a correct tick The smallest element of an image that can be one colour	1	Correct answer only
1(b)	1 mark for each correct answer - Image resolution is the number of pixels wide by pixels high that form the image and screen resolution is the number of pixels wide by pixels high that can be displayed on a screen. - If screen resolution is not as high as image resolution the quality of the image is worsened. // Image resolution can be changed, but screen resolution has a physical limit	2	Accept horizontal and/or vertical for wide and/or high Just number of pixels for either or both is NE Do not accept bits or characters for pixels
1(c)	1 mark for a correct answer For example: - File size - File type - Dimensions of the image // image size - Compression type - Data offset – bytes from the beginning of the file to the actual pixel data	1	Mark the first answer given Do not accept resolution or colour depth, they are given in the question Do not accept a description of a file header or metadata Accept height and/or width for dimensions BOD
1(d)(i)	1 mark for: Two // 2	1	Correct answer only
1(d)(ii)	1 mark for each correct point to max 2 - Run Length Encoding (RLE) can be used - A series of consecutive pixels of the same colour // a sequence of the same colour pixel is stored as a single copy of the colour and the number of consecutive occurrences 1 mark for a suitable reference to the given image - RLE: e.g. R2 // W4 // 2R4W // R2W4	3	Take care that the answers are compressing pixels not anything else Do not accept bits or characters or elements etc. for pixels If no reference to the diagram included in the answer, please check the diagram for annotations

4. M/J 26/P12/Q4/a,b

4(a)	1 mark for: The sound wave is measured 44100 times each second // 44100 samples are taken every second	1 Accept appropriate alternative words for measured including 'sampled' Accept 44.1 thousand instead of 44100
4(b)	1 mark each to max 3 - The resolution of 2 bytes (16 bits) will have more bits per sample ... which will result in a larger file size ... but the digital sound will be more accurate to original ... because a greater range/more exact amplitudes can be recorded ... with smaller quantisation errors	3 Must be clear which resolution is being described Allow reverse arguments e.g. The resolution of 8 bits will have fewer bits per sample ... which will result in a smaller file size etc.

5. M/J 26/P12/Q5/a,b

5(a)	1 mark for each correctly completed value <table><thead><tr><th>denary</th><th>unsigned binary</th><th>hexadecimal</th><th>Binary Coded Decimal (BCD)</th></tr></thead><tbody><tr><td>160</td><td>10100000</td><td>A0</td><td>0001 0110 0000</td></tr><tr><td>289</td><td>100100001</td><td>121</td><td>001010001001</td></tr><tr><td>2486</td><td>1001 1011 0110</td><td>9B6</td><td>0010010010000110</td></tr><tr><td>35123</td><td>1000100100110011</td><td>8933</td><td>00110101000100100011</td></tr></tbody></table>	denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)	160	10100000	A0	0001 0110 0000	289	100100001	121	001010001001	2486	1001 1011 0110	9B6	0010010010000110	35123	1000100100110011	8933	00110101000100100011	4
denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)																			
160	10100000	A0	0001 0110 0000																			
289	100100001	121	001010001001																			
2486	1001 1011 0110	9B6	0010010010000110																			
35123	1000100100110011	8933	00110101000100100011																			
5(b)	1 mark for each correct answer <table><thead><tr><th>statement</th><th>answer</th></tr></thead><tbody><tr><td>The smallest 12-bit two's complement binary number</td><td>1000 0000 0000</td></tr><tr><td>The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.</td><td>24</td></tr><tr><td>The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101</td><td>0101 1001</td></tr></tbody></table>	statement	answer	The smallest 12-bit two's complement binary number	1000 0000 0000	The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24	The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001	3 Note: The first answer must be 12 bits and a binary value Note: The second answer must be in denary.												
statement	answer																					
The smallest 12-bit two's complement binary number	1000 0000 0000																					
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24																					
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001																					

6. M/J 26/P13/Q3/a,b,c(i)

3(a)	1 mark for: Each sample is stored as a 16-bit binary number // there are 2^{16} / 65536 different amplitudes that can be recorded	1 Accept 2 bytes Or any equivalent.
3(b)(i)	1 mark each to max 3 - Image is a series of pixels ... where each pixel has one colour - Each colour has a unique binary number - The binary number/code of the colour of each pixel is stored in sequence	3
3(b)(ii)	1 mark for working 1 mark for answer $2048 * 1536 * 2 * 8 / 1024 / 1024$ 48 MiB	2

3(c)(i)	1 mark each max 2 for image and max 2 for sound Image e.g. • Run-length encoding / RLE • Store each colour and the number of times it appears consecutively Sound e.g. • Reduce amplitude values (dynamic range) to only those sounds used in the actual recording ... • ... reducing sampling resolution without impacting the sound • Record changes in amplitudes instead of each individual amplitude ... • ...each change requires less bits than a new amplitude	4	Accept RLE for sound but do not allow for both parts
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7. M/J 26/P13/Q6/a,b,c,d

6(a)	1 mark for each set of lines from each denary number <table><thead><tr><th>Denary number</th><th>Equivalent unsigned value</th></tr></thead><tbody><tr><td>3120</td><td>Hexadecimal: 121</td></tr><tr><td>289</td><td>BCD: 00010101100001010001</td></tr><tr><td>22516</td><td>Binary: 110000110000</td></tr><tr><td>15851</td><td>Hexadecimal: 57F4</td></tr><tr><td></td><td>Binary: 11110111101011</td></tr><tr><td></td><td>BCD: 0011000100100000</td></tr></tbody></table>	Denary number	Equivalent unsigned value	3120	Hexadecimal: 121	289	BCD: 00010101100001010001	22516	Binary: 110000110000	15851	Hexadecimal: 57F4		Binary: 11110111101011		BCD: 0011000100100000	4	
Denary number	Equivalent unsigned value																
3120	Hexadecimal: 121																
289	BCD: 00010101100001010001																
22516	Binary: 110000110000																
15851	Hexadecimal: 57F4																
	Binary: 11110111101011																
	BCD: 0011000100100000																
6(b)	1 mark for working 1 mark for answer. <pre> 1101 0100 1111 1111 1100 1111 1 1101 0001 1110 1111 1 1 111</pre>	2	No marks for answer if no evidence of using binary addition.														
6(c)	1 mark for correct answer 1713	1															
6(d)	1 mark for correct answer 1000000000000000	1	Last answer														

8. O/N 25/P11/Q1/a,b

1(a)(i)	B3A	1
1(a)(ii)	0001 0000 1000	1

1(a)(iii)	1 mark for the working 1 mark for the correct denary value Working: Method 1: <pre> 1111 1011 1100 Flip the bits 0000 0100 0011 Add 1 0000 0000 0001 + 0000 0100 0100 </pre> Method 2: -2048 + 1024 + 512 + 256 + 128 + 32 + 16 + 8 + 4 // -128 + 32 + 16 + 8 + 4 Denary value: -68	2
1(b)(i)	(1) 00101011	1
1(b)(ii)	1 mark per bullet point, max 2 marks - An overflow error occurs - The answer cannot be represented in the number of bits available // The answer is larger than the maximum positive number that can be stored in the register // The answer is smaller than the most negative number that can be stored in the register	2

9. O/N 25/P11/Q7/d

7(d)(i)	1 mark for the working 1 mark for the correct answer Working: $(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024$ $// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}$ Answer: 1024 kibibytes // 2^{10} kibibytes	2
7(d)(ii)	1 mark per bullet point, max 2 marks - The file will contain metadata - The file will have a header	2
7(d)(iii)	1 mark per bullet point, max 2 marks - Sequences of consecutive identical colours / pixels ... are stored as the colour value and the number of times it occurs consecutively	2

10. O/N 25/P12/Q6/a,b,d

6(a)	1 mark for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	1								
6(b)	1 mark for correct words in the correct order <table><tr><td>Binary Value</td><td>0011 1000</td><td>0011 1100</td><td>0011 1110</td></tr><tr><td>Word</td><td>Science</td><td>is</td><td>Amazing!</td></tr></table>	Binary Value	0011 1000	0011 1100	0011 1110	Word	Science	is	Amazing!	1
Binary Value	0011 1000	0011 1100	0011 1110							
Word	Science	is	Amazing!							
6(d)	1 mark for the working 1 mark for the correct answer Working: (1000 * 2000 * 16) / (8 * 1000 * 1000) // (1000 * 2000 * 2 / (1000 * 1000) Answer: 4 megabytes	2								

11. O/N 25/P12/Q7/a,b,c

7(a)	1 mark for the correct answer 1100 0000	1
7(b)	1 mark for the correct answer 8 // 16 // 32 // 64	1
7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
7(c)(ii)	1 mark for the correct answer 68	1

12. O/N 25/P13/Q1/a,b,c

1(a)	1 mark for each correctly connected change box, max 3 marks	3
1(b)	1 mark per bullet point, max 2 marks - Each character has a unique code - Each character in the text is replaced sequentially / in order by its code	2
1(c)	1 mark per bullet point, max 2 marks - ASCII uses 7 / 8 bits Unicode can use many more / up to 32 bits - Unicode can represent a wider range of characters including different languages	2

13. O/N 25/P13/Q2/a,b,d

2(a)	3 marks for 6 correct ticks 2 mark for 4 or 5 correct ticks 1 mark for 2 or 3 correct ticks <table><tr><th rowspan="2">Example of data</th><th colspan="7">Number of bits</th></tr><tr><th>4</th><th>8</th><th>16</th><th>24</th><th>32</th><th>64</th><th>128</th></tr><tr><td>the hexadecimal value F139</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>16 000 000 unique amplitude values</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>an IPv4 address</td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td></tr><tr><td>256 unique colours</td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>an IPv6 address</td><td></td><td></td><td></td><td></td><td></td><td></td><td>✓</td></tr><tr><td>the denary value 65 000</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr></table>	Example of data	Number of bits							4	8	16	24	32	64	128	the hexadecimal value F139			✓					16 000 000 unique amplitude values				✓				an IPv4 address					✓			256 unique colours		✓						an IPv6 address							✓	the denary value 65 000			✓					3
Example of data	Number of bits																																																																
	4	8	16	24	32	64	128																																																										
the hexadecimal value F139			✓																																																														
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an IPv6 address							✓																																																										
the denary value 65 000			✓																																																														
2(b)	1 mark 1111 1001 0100	1																																																															
2(d)	1 mark E33B	1																																																															

14. O/N 25/P13/Q7/b

7(b)	1 mark for the working 1 mark for the correct answer Working: $(4000 \times 3000 \times 30 \times 16) / (8 \times 1000 \times 1000 \times 1000)$ $// (4000 \times 3000 \times 30 \times 2) / (1000 \times 1000 \times 1000)$ Answer: 0.72 gigabytes	2
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15. M/J 25/P11/Q2/b

2(b)(i)	1 mark each to max 4 e.g. • Video is data-intensive • The file size needs reducing in order to • reduce the amount of bandwidth used • and reduce buffering • This means people are not behind in the conversation • and people with lower bandwidth can still take part	4
2(b)(ii)	No mark for choice but lossy is most appropriate 1 mark each to max 3 e.g. • Reduces file size more than lossless • so significantly less bandwidth / data is needed • so buffering is reduced even more than with lossless • Data can be removed which cannot be seen • reducing quality without impacting experience • for example, because resolution of video can be reduced // sample rate of audio can be reduced	3

16. M/J 25/P11/Q3/a,b

3(a)	1 mark for each correct answer <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The term for the smallest element that makes up an image.</td><td>pixel</td></tr><tr><td>The largest number of different colours that can be represented with a bit depth of 8 bits.</td><td>256 // 2⁸</td></tr><tr><td>The term for the dots per inch (dpi) when an image is displayed.</td><td>Screen resolution</td></tr></table>	Statement	Answer	The term for the smallest element that makes up an image.	pixel	The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸	The term for the dots per inch (dpi) when an image is displayed.	Screen resolution	3
Statement	Answer									
The term for the smallest element that makes up an image.	pixel									
The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸									
The term for the dots per inch (dpi) when an image is displayed.	Screen resolution									
3(b)(i)	1 mark each to max 2 • A wider range of characters can be represented • so characters from more languages can be represented • and symbols such as emojis can be used	2								
3(b)(ii)	1 mark for: 10,094	1								
3(b)(iii)	1 mark for: 8,512	1								

17. M/J 25/P12/Q2/a,b,c

2(a)	1 mark for: 0010 0010 1110 1 mark for: 22E	2
2(b)(i)	1 mark for: -30	1
2(b)(ii)	1 mark for smallest, 1 mark for largest Smallest: 1000 0000 Largest: 0111 1111	2
2(c)	1 mark for application and 1 mark for corresponding justification e.g. - an application that performs financial / banking calculations - because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or - electronic displays - because visual displays only need to show individual digits - because conversion between denary and BCD is straightforward Or - The storage of the date and time in the BIOS of a PC - because conversion between denary and BCD is more straightforward Or - Barcode systems - because conversion between denary and BCD can be accurately completed	2

18. M/J 25/P13/Q1/a,b,c

1(a)(i)	1 mark for correct working: e.g. $2000000 * 16 / (8 * 1000 * 1000)$ 1 mark for answer: 4MB	2
1(a)(ii)	1 mark each Image: - There will be fewer shades of colour available - so the image does not match the original as detail is lost Image file: - Fewer bits are used to store each pixel - so less data is stored, therefore the file size is reduced	4

1(b)	1 mark for each correct term:	3																
<table><tr><th>Description</th><th>Sound term</th></tr><tr><td>the number of times the amplitude is measured per time interval</td><td>Sampling rate</td></tr><tr><td>the number of bits used to store each amplitude measurement</td><td>Sampling resolution</td></tr><tr><td>the type of sound wave before it is recorded by a computer</td><td>Analogue</td></tr></table>		Description	Sound term	the number of times the amplitude is measured per time interval	Sampling rate	the number of bits used to store each amplitude measurement	Sampling resolution	the type of sound wave before it is recorded by a computer	Analogue									
Description	Sound term																	
the number of times the amplitude is measured per time interval	Sampling rate																	
the number of bits used to store each amplitude measurement	Sampling resolution																	
the type of sound wave before it is recorded by a computer	Analogue																	
1(c)(i)	1 mark each to max 2 8 / 16 / 32 / bits per character - Represents $2^8 / 2^{16}$ / etc. characters - Represents every language and other characters such as emojis	2																
1(c)(ii)	1 mark for each correctly completed space <table><tr><th>Character</th><th>Denary</th><th>8–bit Binary</th><th>Hexadecimal</th></tr><tr><td>!</td><td>33</td><td>0010 0001</td><td>21</td></tr><tr><td>L</td><td>76</td><td>0100 1100</td><td>4C</td></tr><tr><td>ü</td><td>252</td><td>1111 1100</td><td>FC</td></tr></table>	Character	Denary	8–bit Binary	Hexadecimal	!	33	0010 0001	21	L	76	0100 1100	4C	ü	252	1111 1100	FC	3
Character	Denary	8–bit Binary	Hexadecimal															
!	33	0010 0001	21															
L	76	0100 1100	4C															
ü	252	1111 1100	FC															

19. M/J 25/P13/Q7/b,c

7(b)	1 mark for $ \begin{array}{r} 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ +_1 1_1\ 0_1\ 1_1\ 1\ 0\ 0\ 0_1\ 1 \\ \hline (1)\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 0 \end{array} $ Answer: (1) 1010 0110	1
7(c)	1 mark for showing binary subtraction (any method) Direct subtraction: $ \begin{array}{r} 0\ 1\ 1\ 10\ 10\ 11\ 10\ 0 \\ -0\ 0\ 0_1\ 1_1\ 1_1\ 1_1\ 1\ 0 \\ \hline 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ Adding the two's complement: $ \begin{array}{r} 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0 \\ +_1 1_1\ 1_1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline (1)\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ 1 mark for answer 0100 0110	2

20. O/N 24/P11/Q1/a,b,c,d

1(a)	1 mark for: - A tebibyte = 1024 gibibytes / 1 048 576 kibibytes / 2^{40} bytes whereas a gigabyte = 1000 megabytes / 1 000 000 kilobytes / 10^9 bytes - Tebi is binary prefix giga is denary prefix	1
1(b)(i)	1 mark for: C67	1
1(b)(ii)	1 mark for: -1641	1
1(b)(iii)	1 mark for: 573	1
1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 0100 1010 + <u>1110 1001</u> (1)0011 0011 Direct subtraction: 0100 1010 - <u>0001 0111</u> 0011 0011 Answer: 0011 0011	2
1(d)	1 mark from: - The answer cannot be represented in the number of bits available - The answer is larger than the maximum positive number that can be stored in the register - The answer is smaller than the most negative number that can be stored in the register	1

21. O/N 24/P11/Q4/b

4(b)(i)	1 mark for each bullet point (max 3) e.g. • Less of the teacher's storage space is used • ... so more student work can be stored • Transmission time is reduced • ... so the teacher does not have to wait as long for it to arrive • Bandwidth usage is reduced • ... so other transmissions are not adversely affected • Less data allowance is used on the teacher's email system	3
4(b)(ii)	1 mark for each bullet point (max 3) • Run-Length Encoding // RLE • Repeated sequences of the same characters are replaced by • ... a single copy of the character • ... and a count of the number of characters	3

22. O/N 24/P12/Q7/a,b

7(a)(i)	1 mark for each correct definition Colour depth: • the number of bits used to represent a colour // the number of colours that can be represented in an image File header: • stores data about the image file / metadata	2
7(a)(ii)	1 mark for each correct explanation Image quality: • Decreasing resolution means details within the image are lost because there are fewer pixels // Increasing resolution means the image is more detailed because there are more pixels File size: • Decreasing the resolution will decrease the file size because there are fewer pixels therefore less data // Increasing the resolution will increase the file size because there are more pixels therefore more data	2
7(a)(iii)	1 mark for correct method For example: • Run-Length Encoding	1

7(b)	1 mark for each correct definition Property: - an attribute of a drawing object // data about a shape // defines one aspect of the appearance of a drawing object Drawing list: - all the drawing objects/shapes in an image // stores the commands/descriptions / mathematical equations required to draw each object	2
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23. O/N 24/P13/Q6/a,b,c

6(a)	1 mark for identification 1 mark for matching description e.g. - ASCII 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet - UNICODE 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	2
6(b)(i)	1 mark for: 256 // 2^8	1
6(b)(ii)	1 mark for: Increased file size	1
6(b)(iii)	1 mark for each bullet point (max 2) e.g. - The change may not be noticeable // Data removed is usually not noticed by the human eye ... for example, changes in shade/detail - It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	2
6(c)(i)	1 mark for each bullet point (max 2) - Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals - Each sample/reading/measurement is given the binary number and stored in sequence	2

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. • Decrease sample rate • ... fewer samples/readings/measurements stored per second // fewer bits per second stored • Decrease sample resolution • ... fewer bits per sample/reading/measurement // each sample has fewer bits • Sound outside of set/human hearing range is removed • ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
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24. O/N 24/P13/Q8/a,b

8(a)	1 mark for: Denary value: 8107	1
8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3

25. M/J 24/P11/Q2/d

2(d)	1 mark each to max 3 : • Data transmission to user's smartphone will take longer • ... because there is more data to transmit • The secondary storage device will fill faster • ... fewer videos will be able to be stored long-term // videos are overwritten more often	3
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26. M/J 24/P11/Q7

7	1 mark each: - Working – carried values clearly indicated - Correct answer 0001 1000 - Overflow clearly indicated as overflow Example: <pre> 1 0 0 1 1 1 1 0 0 1 1 0 0 0 0 1 + 0 0 0 1 1 0 0 1 ----- (1) 0 0 0 1 1 0 0 0 1 1 1 1 1 1 1(carries) </pre>	3
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27. M/J 24/P12/Q2/d

2(d)(i)	1 mark each: - Image is made of pixels and each pixel has one colour - Each colour has a unique binary code Code for the colour of each pixel is stored in sequence	3
2(d)(ii)	1 mark each to max 2: - List of objects in the drawing - A list that stores the command/description/equation required to draw each object - Properties of each object e.g. the fill colour, line weight/colour	2
2(d)(iii)	1 mark each to max 2: - Dedicated connection to the headset // not sharing bandwidth - Already fast connection that can transmit the data without slowing - Video may already be a small file size and does not need further reduction - Video is not saved so storage is not an issue in the headset	2

28. M/J 24/P12/Q7/a,b,c

7(a)	1 mark for: 3300 kibibytes	1
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7(b)	1 mark each: - Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 - Subtraction method - converting 10 to – 10 and adding // direct subtraction correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 - 0 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 </pre>	3
7(c)	1 mark for working: 1100 0000 1111 // 2048 + 1024 + 8 + 4 + 2 + 1 // (12 * 16²) + 15 // (12 * 16 * 16) + 15 // 3072 + 15 1 mark for correct answer: 3087	2

29. M/J 24/P13/Q1/a,b,c,d

1(a)	1 mark for each correct answer: - binary 3072 - denary/decimal 2000 A kibibyte has a binary prefix. Three kibibytes is the same as 3072 bytes. A megabyte has a decimal/denary prefix. Two terabytes is the same as 2000 gigabytes.	4
1(b)	1 mark for correct answer: F1	1

1(c)	1 mark for a correct answer: The answer is too long to be represented in the same number of bits as the binary numbers being added	1								
1(d)(i)	1 mark for all 3 answers correct: <table><tr><th>Character set</th><th>Number of bits</th></tr><tr><td>ASCII</td><td>7</td></tr><tr><td>extended ASCII</td><td>8</td></tr><tr><td>Unicode</td><td>16/32</td></tr></table>	Character set	Number of bits	ASCII	7	extended ASCII	8	Unicode	16/32	1
Character set	Number of bits									
ASCII	7									
extended ASCII	8									
Unicode	16/32									
1(d)(ii)	1 mark each: - Each character has a unique binary code - The binary code for each character is stored in sequence	2								

30. M/J 24/P13/Q2/a,b,c

2(a)	1 mark for working: $4000 * 3000 * 4$ 1 mark for correct answer: 48MB	2
2(b)(i)	1 mark each to max 3 : - The file takes less storage space on the web server than if lossless compression was used - The file is faster to upload/download to/from the server than if lossless compression was used - The file uses less bandwidth to transmit than if lossless compression was used - The file consumes less data allowance than if lossless compression was used	3
2(b)(ii)	1 mark each: - Identifies consecutive repeating pixels of the same colour - Stores the colour /pattern and the number of times it repeats	2
2(c)	1 mark each to max 2 : - Colour/bit depth - Image resolution	2

31. O/N 23/P11/Q1/a,b,c

1(a)	2 marks for all 3 lines correct 1 mark for 1 line correct <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>drawing list</td><td>a component created using a formula</td></tr><tr><td>drawing object</td><td>defines one characteristic of a component</td></tr><tr><td>property</td><td>data required to create all components in the graphic</td></tr></tbody></table>	Term	Description	drawing list	a component created using a formula	drawing object	defines one characteristic of a component	property	data required to create all components in the graphic	2
Term	Description									
drawing list	a component created using a formula									
drawing object	defines one characteristic of a component									
property	data required to create all components in the graphic									
1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3								
1(c)	1 mark for each bullet point (max 2) - Reduced bandwidth usage when transmitting the message - Reduced transmission time from email client to email server - Reduced storage space on the email - Email accounts often have a maximum size for an attachment	2								

32. O/N 23/P12/Q3/a,b,c

3(a)	1 mark for: Either - kibibyte = 1,024 bytes // 2^{10} bytes and megabyte = 1000 kilobytes // 1 000 000 bytes // 10^3 kilobytes // 10^6 bytes Or - kibi is binary prefix and mega is denary prefix	1
3(b)(i)	1111 0011 1100	1
3(b)(ii)	865	1
3(b)(iii)	470	1

3(c)	1 mark for a correct application 1 mark for a corresponding justification • An application that performs financial / banking calculations • ... because it is difficult to represent decimal values exactly in normal binary and financial transactions use only two decimal places and must be accurate, no accumulating errors Or • Electronic displays, e.g. calculators, digital clocks • ... because visual displays only need to show individual digits • ... because conversion between denary and BCD is easier Or • The storage of the date and time in the BIOS of a PC • ... because conversion with denary is easier	2
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33. O/N 23/P12/Q6/a,b,c

6(a)	1 mark for each bullet point (max 3) Lossy compression (ticked) • Loss of quality will not be noticed • Needs to be viewed in real time so less bandwidth needed if file size smaller • Smaller file sizes will reduce buffering so the video will play more smoothly • Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) • Original recording may not have been made in high resolution • Could be streaming to high bandwidth devices • The reduction in the file size is sufficient for the receiving device • Viewers do not want any loss of quality	3
6(b)	1 mark for each bullet point (max 3) Increase sampling resolution • ... the number of bits used for each sample is increased • ... there will be more values available to represent each sample // more amplitudes can be represented • ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note • ... quantisation errors are reduced • ... the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution • ... the number of bits used for each sample is decreased • ... there will be fewer values available to represent each sample // fewer amplitudes can be stored • ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note • ... quantisation errors are increased • ... the digital soundwave is less like the original analogue soundwave	3

6(c)	1 mark for answer, 1 mark for working Answer: 2.5 mebibytes Working: $(2048 \times 1024 \times 10) / (8 \times 1024 \times 1024)$	2
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34. O/N 23/P13/Q1/a,b

1(a)	1 mark for: data values that are continuously changing // variable // any value	1								
1(b)	1 mark for 1 correct line 2 marks for 3 correct lines <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>sampling</td><td>the number of samples taken per second</td></tr><tr><td>sampling rate</td><td>taking measurements at regular intervals and storing the values</td></tr><tr><td>sampling resolution</td><td>the number of bits used to store each sample</td></tr></tbody></table>	Term	Description	sampling	the number of samples taken per second	sampling rate	taking measurements at regular intervals and storing the values	sampling resolution	the number of bits used to store each sample	2
Term	Description									
sampling	the number of samples taken per second									
sampling rate	taking measurements at regular intervals and storing the values									
sampling resolution	the number of bits used to store each sample									

35. O/N 23/P13/Q2/a,b

2(a)	1 mark for each bullet point: - more pixels can be stored / are available - the image is sharper / less pixelated	2
2(b)	1 mark for answer, 1 mark for working Answer: 4096 kibibytes Working: File size = $(2048 \times 1024 \times 16) / (8 \times 1024)$	2

36. O/N 23/P13/Q5/b(i)

5(b)(i)	1 mark for each bullet point (max 2). - To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place - To reduce the amount of storage space used on the web server // user's device	2
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37. M/J 23/P11/Q1/a,b,c

1(a)	1 mark for each correct definition <table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>	Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image	3
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									
1(b)(i)	1 mark each Examples: • Confirmation that it is a bitmap // file type • Compression type • Location/offset of data within the file • Dimensions e.g.100 × 100 pixels	2								
1(b)(ii)	1 mark for working; 1 mark for answer • Working: (1500 * 3000 * 8) / 1000 / 1000 • Answer: 36 MB	2								
1(c)(i)	1 mark each Examples: • The customers will be able to download the photographs in less time • ...and they will take less of the customer's bandwidth • The photographs will take up less space on the customer's storage medium • ...therefore the customers can store more images • ...and will have more space for other files	3								

1(c)(ii)	1 mark each to max 2 for explanation; 1 mark for an image related example - RLE stores a colour and the number of times it occurs consecutively - An image may not have many sequences of the same colour - It would need to store each colour and then the count/number 1 which adds data Example: - Red-Green-Blue would become Red 1 Green 1 Blue 1	3
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38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

3(d)(i)	1 mark from - Kibibyte is 1024 bytes and kilobyte is 1000 bytes - Kibibyte is binary prefix and kilobyte is denary prefix	1
3(d)(ii)	1001 0110 0100	1
3(d)(iii)	F2	1
3(d)(iv)	Smallest: 10000000 Largest: 01111111	2
3(d)(v)	1 mark for working 1 mark for answer <pre> 1 0 1 1 0 0 0 0 + 0 0 0 1 1 0 1 1 1 1 0 0 1 0 1 1 1 1 </pre>	2

39. M/J 23/P12/Q4/a,b,c

4(a)	2^{16} // 65536	1
4(b)	1 mark for working; 1 mark for answer - Working: +120 = 0111 1000 - Answer: 1000 0111	2
4(c)	1 mark for working; 1 mark for answer - Working: $A04 = (10 * 16^2) + 4$ // $A04 = (10 * 256) + 4$ // $A04 = 1010 0000 0100$ - Answer: 2564	2

40. M/J 23/P13/Q3/c

3(c)(i)	1 mark each • The amplitude is recorded a set number of times a second • Each (instance of an) amplitude is given a corresponding binary number • The binary number (of each amplitude) is saved in sequence	3
3(c)(ii)	1 mark each; max 2 for rate and max 2 for resolution Sampling rate • There are smaller 'gaps' in the sound wave // sound is recorded more often • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller Sampling resolution • There are more bits per sample // a wider range of amplitudes can be stored • Each binary amplitude/note (in the digital recording) is closer to the analogue amplitude/note • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller	4

41. M/J 23/P13/Q7/a

7(a)	1 mark for each correct line	3
Description		Denary value
The smallest integer that can be represented in 8-bit two's complement.		-127
		127
The largest integer that can be represented in 8-bit two's complement.		-255
		-128
The largest unsigned integer that can be represented in 8 bits.		-256
		256
		128
		255

42. O/N 22/P11/Q1/a,b,c,d

1(a)(i)	39	1
1(a)(ii)	27	1
1(a)(iii)	-25	1

1(b)	1 mark for working, 1 mark for answer (0011 1110) Working using borrowing: <pre> 1 10 10 1 0 10 0 0 10 10 1 0 1 1 0 0 1 1 0 1 1 1 0 1 0 1 - ----- 0 0 1 1 1 1 1 0 </pre> Working using two's complement: <pre> 1 0 1 1 0 0 1 1 Two's complement 1 0 0 0 1 0 1 1 + ----- (1) 0 0 1 1 1 1 1 0 1 1 </pre>	2
1(c)	1 mark for similarity, 2 marks for differences Similarity (max 1): • both can use 8 bits • both represent each character using a unique code • Unicode will contain all the characters that ASCII contains // ASCII is a subset of Unicode Differences (max 2): • Unicode can go up to 32 bits per character whereas ASCII is 7 or 8 bits Unicode can represent a wider range of characters than ASCII • different languages are represented using Unicode, ASCII is only for one language	3
1(d)(i)	the number of samples taken per unit time / per second	1
1(d)(ii)	1 mark for each bullet point (max 2): • increases the number of bits per sample // larger range of values • which means that the file size increases • makes the sound file more accurate //digital waveform closer to original (analogue) waveform • smaller quantisation errors	2

43. O/N 22/P11/Q7/c

7(c)	1 mark for each bullet point (max 3): • lossless compression • Run Length Encoding • repeated sequences of <u>characters</u> are replaced by • ... a single copy of the character • ... and a counter of the number of characters	3
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44. O/N 22/P12/Q2/a,b

2(a)(i)	−106	1
2(a)(ii)	96	1
2(a)(iii)	1 mark for each bullet point: • 149 decimal • 0001 0100 1001	2
2(b)	1101 0010	1

45. O/N 22/P12/Q6/b

6(b)(i)	1 mark for each bullet point (max 2): • improves the accuracy of the sound file • ... because (digital) waveform more closely resembles the analogue waveform • quantization errors are reduced • increases the amount of detail stored	2
6(b)(ii)	1 mark for each bullet point: • decreases the file size of the sound file • ... because fewer bits are used to store each sample	2

46. O/N 22/P12/Q8/a,b,c

8(a)	1 mark one or two correct row(s). 2 marks for all three correct rows.				2
		Action	Increases the file size	Decreases the file size	
		Change the colour depth of the image file to 16 bits per pixel.		✓	
		Change the screen resolution to 1366 × 768 pixels.			✓
		Change the colour of the rectangle from black to red.			✓

8(b)	1 mark for each bullet point (max 2). For example: • can be enlarged without pixelation / loss of quality • individual components of the image can be edited • generally a smaller file size	2								
8(c)(i)	1 mark for each correct underlined part: <table><tr><td>Uncompressed sound</td><td>RLE compressed sound</td></tr><tr><td>EA F1 F1 F2 F2 F2 EA</td><td>1EA 2F1 3F2 1EA</td></tr><tr><td><u>AB AB FF FF 1D 67</u></td><td>2AB 2FF 11D 167</td></tr><tr><td>32 32 80 81 81</td><td><u>232 180 281</u></td></tr></table>	Uncompressed sound	RLE compressed sound	EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA	<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167	32 32 80 81 81	<u>232 180 281</u>	2
Uncompressed sound	RLE compressed sound									
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA									
<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167									
32 32 80 81 81	<u>232 180 281</u>									
8(c)(ii)	1 mark for each bullet point: • all the data is required // no data can be lost • ... otherwise text file will be corrupted / not make sense	2								

47. O/N 22/P13/Q1/a,b

1(a)	1 mark for one or two correct ticks, 2 marks for three correct ticks.	2																
	<table><tr><th>Action</th><th>Accuracy increases</th><th>Accuracy decreases</th><th>Accuracy does not change</th></tr><tr><td>Change the sampling rate from 40 kHz to 60 kHz.</td><td>✓</td><td></td><td></td></tr><tr><td>Change the duration of the recording from 20 minutes to 40 minutes.</td><td></td><td></td><td>✓</td></tr><tr><td>Change the sampling resolution from 24 bits to 16 bits.</td><td></td><td>✓</td><td></td></tr></table>	Action	Accuracy increases	Accuracy decreases	Accuracy does not change	Change the sampling rate from 40 kHz to 60 kHz.	✓			Change the duration of the recording from 20 minutes to 40 minutes.			✓	Change the sampling resolution from 24 bits to 16 bits.		✓		
Action	Accuracy increases	Accuracy decreases	Accuracy does not change															
Change the sampling rate from 40 kHz to 60 kHz.	✓																	
Change the duration of the recording from 20 minutes to 40 minutes.			✓															
Change the sampling resolution from 24 bits to 16 bits.		✓																
1(b)	1 mark for answer; 1 mark for working. Working: Size = 50KHz * (20 × 60) * 16 bits = 50 000 * 1200 * 16 bits // 50 000 *1200 * 2 bytes = 960 000 000 bits = 120 000 000 bytes = 120 000 kilobytes = 120 megabytes Answer = 120 megabytes	2																

48. O/N 22/P13/Q2/e

2(e)	1 mark for each correctly completed term; The bit depth of a bitmap image is the number of bits that are used to store each pixel. Metadata about the image is stored in the header of the file.	2
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49. O/N 22/P13/Q9/a,b

9(a)(i)	93	1
9(a)(ii)	147	1
9(b)	1 mark for each correct benefit (max 2) Examples: • straightforward to convert to / from BCD and denary • ... so it is less complex to encode and decode for programmers • easier for digital equipment use BCD to display output information • can represent monetary values exactly	2

50. O/N 21/P11/Q1/a,b,c

1(a)	1 mark for each correct line	5
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1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2
1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

51. O/N 21/P11/Q7/a,b

7(a)(i)	1 mark per bullet point • Smaller time gaps between the samples • Makes the digital sound wave more accurate • Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point • More samples/data are taken/recorded • ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2

52. O/N 21/P12/Q4/a,b,c,d,e

4(a)	205	1
4(b)	-51	1
4(c)	CD	1

4(d)	1 mark for: The denary value in each group of 4 bits is greater than 9 // the denary value in each nibble is greater than 9	1
4(e)(i)	1 mark for working, 1 mark for answer <pre> 0011 1101 +0010 1101 ----- 0110 1010 111 1 1 </pre>	2
4(e)(ii)	1 mark for working, 1 mark for answer <pre> 0011 1101 +1101 0011 (two's complement) ----- 0001 0000 1111 111 </pre>	2

53. O/N 21/P12/Q5/a,b

5(a)	1 mark for each description, 1 mark for each valid example <table border="1"> <thead> <tr> <th>Term</th><th>Description</th><th>Example from logo</th></tr> </thead> <tbody> <tr> <td>Property</td><td>data about the shapes // defines one aspect of the appearance of the drawing object</td><td>e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle</td></tr> <tr> <td>Drawing list</td><td>the list of shapes involved in an image // a list that stores the command/description required to draw each object</td><td>e.g. triangle // capital letter R // rectangle // line</td></tr> </tbody> </table>	Term	Description	Example from logo	Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle	Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line	4
Term	Description	Example from logo									
Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle									
Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line									
5(b)(i)	1 mark for each bullet point to max 2 for each difference • Bitmap made up of pixels // bitmap is made of colours stored for individual pixels • Vector graphic store a set of instructions about how to draw the shape • When bitmap is enlarged the pixels get bigger and it pixelates • When vector is enlarged it is recalculated and does not pixelate • Bitmap files are usually bigger than vector graphics files because of the need to store data about each pixel • Vector graphics have smaller file size because they contain just the instructions to create the shapes • Bitmap images can be compressed with significant reduction in file size • Vector graphic images do not compress well because of little redundant data	4									

5(b)(ii)	1 mark for each bullet point to max 2 for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	4
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54. O/N 21/P13/Q1/a,b,c

1(a)	1 mark for each correct line	5
1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2

1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

55. O/N 21/P13/Q7/a,b

7(a)(i)	1 mark per bullet point - Smaller time gaps between the samples - Makes the digital sound wave more accurate - Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point - More samples/data are taken/recorded ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point - Reduces the file size - Faster to transmit/download - Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. - Reduce amplitude to only the range used ... limited amplitudes mean fewer bits per sample - Run-length-encoding ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats - Record the changes instead of the actual sounds	2

56. M/J 21/P11/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: - A single square of one colour - The smallest addressable element in an image File header: - Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: $1024 \times 512 = 524\,288$ pixels/bytes $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3

1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer $\begin{array}{r} 0010 \ 0011 \\ + \ 0000 \ 1111 \\ \hline 0011 \ 0010 \\ 1 \ 111 \end{array}$	3
1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 $\begin{array}{r} 10 = 0000 \ 1010 \\ -10 = 1111 \ 0110 \\ \\ 0010 \ 0011 \\ + \ 1111 \ 0110 \\ \hline 0001 \ 1001 \\ 11 \ 11 \end{array}$	3

57. M/J 21/P12/Q4/b(i)

4(b)(i)	102	1
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58. M/J 21/P12/Q6/a,b,c

6(a)	1 mark for each correct answer ASCII = $128 // 2^7$ Extended ASCII = $256 // 2^8$	2
6(b)	1 mark per bullet point to max 2 • Each character has its own unique code • Each character in the word is replaced by its code • The codes are stored in the order in the word	2
6(c)(i)	31	1
6(c)(ii)	53	1

59. M/J 21/P13/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: • $1024 \times 512 = 524\,288$ pixels/bytes • $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
1(b)	1 mark for naming method, 1 mark per description to max 2 • Run-length encoding • Replace sequences of the same colour pixel • ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer <pre> 0010 0011 + 0000 1111 ----- 0011 0010 1 111 </pre>	3
1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 <pre> 10 = 0000 1010 -10 = 1111 0110 0010 0011 + 1111 0110 ----- 0001 1001 11 11 </pre>	3